SESSION 7 – LAB REPORT

Tip Calculator Console Program

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1. Aim

To develop a Kotlin console program that calculates the tip amount for a given bill using functions, parameters, conditional logic, and numeric formatting. The program also demonstrates optional rounding of the tip amount.

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2. Problem Statement

Build a console-based Tip Calculator in Kotlin that computes the tip based on bill amount and tip percentage. The program should optionally round up the tip using mathematical functions and display the result formatted to two decimal places.

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3. Program Code

```
import java.util.Locale import
kotlin.math.ceil

// Function to calculate tip
fun calculateTip(amount: Double, tipPercent: Int, roundUp: Boolean): Double
{
    var tip = amount * tipPercent / 100.0

    if (roundUp) {
tip = ceil(tip)
    }
return tip
}

// Function to format amount to two decimal places
fun formatAmount(amount: Double): String {
return String.format(Locale.US, "%.2f", amount)
}  fun
main() {
// Scenario
```

```
1     val
bill1 =
50.00
    val tip1 = calculateTip(bill1, 15, false)     println("Bill:
${formatAmount(bill1)} | Tip percent: 15% | Round up:
false | Tip: ${formatAmount(tip1)}")

    // Scenario 2
val bill2 = 75.20
    val tip2 = calculateTip(bill2, 18, true)     println("Bill:
${formatAmount(bill2)} | Tip percent: 18% | Round up:
true  | Tip: ${formatAmount(tip2)}")

    // Scenario 3
val bill3 = 100.00
    val tip3 = calculateTip(bill3, 20, false)     println("Bill:
${formatAmount(bill3)} | Tip percent: 20% | Round up:
false | Tip: ${formatAmount(tip3)}")
                                                                                }
```

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4. Sample Output

```
Bill: 50.00 | Tip percent: 15% | Round up: false | Tip: 7.50
Bill: 75.20 | Tip percent: 18% | Round up: true  | Tip: 14.00
              Bill: 100.00 | Tip percent: 20% | Round up: false | Tip: 20.00
```

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5. Explanation

The program uses a pure function `calculateTip()` to compute the tip based on the bill amount and tip percentage. If the roundUp parameter is true, the `ceil()` function is used to round the tip to the next whole number. The `formatAmount()` function ensures that the output is formatted to two decimal places using `String.format()` with `Locale.US`. The `main()` function demonstrates three different scenarios to validate the correctness of the logic.

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6. Concepts Used

- Kotlin Functions
- Parameter Passing
- Conditional Statements
- Mathematical Functions (`ceil`)
- Double Data Type
- String Formatting using Locale
- Clean Code Structure

7. Conclusion

This lab strengthened my understanding of numeric operations and conditional logic in Kotlin. I learned how to write reusable pure functions and properly format output values. The program demonstrates how structured logic and functions can be used to build real-world calculation-based applications.